

Gym Member Attendance Monitoring System

ASP.NET Core MVC Laboratory Exercise Specification

Objective

Develop an ASP.NET Core MVC application to monitor gym member attendance using Model Binding, Cookie Authentication, Data Annotation Validation, Repository Pattern (in-memory), Razor Views, Bootstrap, CRUD, and Search.

Business Scenario

Gym staff record member visits, monitor who is currently inside the facility, and record checkout times.

Learning Objectives

MVC, Model Binding, DTOs, Validation, Cookie Authentication, Authorization, Repository Pattern, CRUD, Search, Bootstrap.

Functional Requirements

Register User, Login, Logout, Register Member Visit, View Attendance List, Edit Visit, View Details, Record Checkout, Search.

Visit Fields

Visit Number, Member ID, First Name, Last Name, Membership Type, Contact Number, Visit Date, Time In, Time Out, Workout Purpose, Notes.

Monitoring

Display Visit Number, Member, Membership, Time In, Time Out, Status, Actions.

Status

Inside Gym, Checked Out.

Validation

Use Required, StringLength, Phone, Display, DataType and validate ModelState.

Authentication

Public: Login/Register. Protected: Attendance pages using Cookie Authentication and [Authorize].

Entities

User: Id, FirstName, LastName, Email, Username, Password. MemberVisit: Id, VisitNumber, MemberId, FirstName, LastName, MembershipType, ContactNumber, VisitDate, TimeIn, TimeOut, Status, WorkoutPurpose, Notes.

Repositories

UserRepository and MemberVisitRepository using static List.

Controllers & Views

Controllers: HomeController, AccountController, MemberVisitController. Views: Login, Register, Index, Create, Edit, Details, CheckOut.

Submission

Complete MVC application with authentication, CRUD, validation, search, Bootstrap UI, repository pattern, and clean code.